



R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Course Code:

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Structural Dynamics	MTSE-2001	3L-1T-0P	4

After studying this subject, the students will be able to:

CO1: Understand the concept of free and forced vibrations.

CO2: Analyze the Single degree of freedom Systems.

CO3: Analyze the Multi degree of freedom Systems.

CO4: Evaluate the responses by numerical methods.

CO5: Analyze various continuous systems.

Course Contents:

Unit 1: Free and Forced vibrations

Introduction to Structural Dynamics, Time Dependent Problems, Degrees of Freedom, Simple Harmonic Motion, Structural Vibration, Damping, Equations of Motion and Natural Frequency, Modelling of SDOF Structures, D'Alemberts principles, Undamped and damped Free Vibration Response, Logarithmic Decrement, Undamped and damped harmonic and periodic excitation.

Unit 2: Single degree of freedom Systems

Resonant Frequency & Half Power Band Width, Force Transmission and Vibration Isolation, Vibration Measuring Instruments, Equivalent viscous damping, Duhamel's integral, Response to arbitrary and Impulse excitations, Fourier series, Frequency response function, Energy methods, Rayleigh's method.

Unit 3: Multi degree of freedom Systems

Matrix formulation, stiffness and flexibility influence coefficients, eigen value problem, normal modes and their properties. Matrix iteration technique for eigen value, and eigen vectors, Free and forced vibration by modal analysis, orthogonality of modes, Modal Superposition Method.

Unit 4: Numerical Methods and Response Spectrum

Numerical Methods, Methods Based on Interpolation of Excitation, Central Difference Method, Newmark's Method, Response Spectrum concept and characteristics, Special Cases of Response Spectrum, Development of Tripartite Plot.

Unit 5: Continuous Systems

Equations of Motion (for String, Bar, Beam), Axial vibration of bar, Vibration of beams, dynamic matrices for flexural, axial and torsional effects, approximate methods of Rayleigh-Ritz, Galerkin etc.

Reference Books

1. A. K. Chopra, Dynamics of Structures: Theory and Applications to Earthquake Engineering, Prentice Hall of India
2. R. W. Clough, J. Penzien, Dynamics of Structures, McGraw-Hill
3. J. L. Humar, Dynamics of Structures, CRC Press
4. L. Meirovitch, Elements of Vibration Analysis, McGraw-Hill.
5. M. Paz, Structural Dynamics, CBS Publ. New Delhi.
6. S. Timoshenko., Advanced Dynamics, McGraw Hill Book Co, NY.

Course outcomes:

- After completion of the course the students will have the knowledge of vibration analysis of systems/structures with different degrees of freedom and they know the method of damping the systems.



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First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	FEM in Structural Engineering	MTSE-2002	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the concept of finite element method in structural engineering.

CO2: Understand the properties and formulation of various finite elements.

CO3: Analyze the various structural members such as springs, truss/bars, beams, plane, grid and space frames

CO4: Analyze the two-dimensional, three-dimensional and axisymmetric finite elements.

CO5: Analyze of plates, elastic stability, dynamic numerical using FEM.

Course Contents

Unit 1: Introduction to Finite Element Method

Introduction, Basic concepts of finite element analysis, Introduction to elasticity, Virtual work, Variational principle, Galerkin method, Basic finite element concepts, Basic ideas in a finite element solution, General finite element solution procedure, Stiffness matrix and Boundary conditions.

Unit 2: Element Properties

Natural coordinates, Interpolation Polynomials, their selection and derivation in terms of global and local in Cartesian and natural coordinate systems, Convergence requirements, Assembly of Element matrices, Isoparametric Formulation, Lagrange and Hermite interpolation functions, Numerical Integration

Unit 3: General Procedure of Finite Element Method

Stiffness of spring, Analysis of spring, Stiffness of Truss/bar Members, Analysis of Truss, Stiffness of Beam Members, Finite Element Analysis of Continuous Beam, Plane Frame Analysis, Analysis of Grid and Space Frame.

Unit 4: FEM for Two and Three Dimensional Solids

Two Dimensional Solids: Constant Strain Triangle, Linear Strain Triangle, Rectangular Elements - Numerical Evaluation of Element Stiffness -Computation of Stresses, Geometric Nonlinearity and Static Condensation

Axisymmetric Element: Finite Element Formulation of Axisymmetric Element Finite Element Formulation for 3 Dimensional Elements, Examples

Unit 5: Applications of FEM

Plate Bending Problems, Finite Elements for Elastic Stability, Dynamic Analysis, Examples

Reference Books

1. Reddy J. N., An Introduction to Finite Element Method, McGraw-Hill, Intl. Student Edition, 1985.
2. Zienkiewics, The finite element method, Basic formulation and linear problems, Vol.1, 4th Edition, McGraw-Hill, Book Co., 1987
3. Rao S. S, The Finite Element Method in Engineering, Pergaman Press, 2003.
4. Desai C.S. and. Abel J.F, Introduction to the Finite Element Method, Affiliated East West Press, 1972.
5. Cook R. D., Concepts and Applications of Finite Element Analysis, Wiley and Sons, 1989.
6. Buchanan, Finite element Analysis (Schaum Outline S), TMH
7. Chandrupatla T. R., and Belegundu A. D., Introduction to Finite Element in Engineering, Third Edition, Prentice Hall, India, 2007.
8. Krishnamoorthy C. S., Finite Element Analysis Theory and Programming, Tata McGraw Hill Education, 1994



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Course Content & Grade

9. Hutton D. V., Fundamentals of Finite Element Analysis, Tata McGraw Hill Publishing Company Limited, New Delhi, 2005
10. Logan D. L., A First Course in Finite Element Method, Cengage Learning, 2012.
11. Bhavikatti S. S, Finite Element Analysis, New Age International Publishers, 2007.



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Design of Earthquake Resistant Structures	MTSE-2003	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the concepts of Engineering Seismology.

CO2: Understand the concepts Response Spectrum.

CO3: Analyze and design various earthquake resistant members.

CO4: Design earthquake resistant structures for ductile detailing.

CO5: Understand the concept of various seismic control devices.

Course Contents

Unit 1: Engineering Seismology

Introduction to engineering seismology, Geological and tectonic features of India, Origin and propagation of seismic waves, Earthquake measurement parameters, Characteristics of earthquake and its quantification- Magnitude and Intensity scales, Seismic instruments. Seismic zoning map of India.

Unit 2: Response Spectrum

Response history and strong motion characteristics. Response Spectrum- elastic and inelastic response spectra, tripartite (D-V-A) response spectrum, use of response spectrum in earthquake resistant design. Computation of seismic forces in multi-storeyed buildings - using procedures as per codal provisions.

Unit 3: Aseismic Structural Modelling

Structural configuration for earthquake resistant design, Concept of plan irregularities and vertical irregularities, Soft storey, Torsion in buildings, Design as per relevant codal provisions, Effect of infill masonry walls on frames, modeling concepts of infill masonry walls. Behavior of masonry buildings during earthquakes, failure patterns, strength of masonry in shear and flexure, Slenderness concept of masonry walls,

Unit 4 Design of Structure for Earthquake Resistance

Seismic design philosophy, Load combinations, Ductility and energy absorption in buildings. confinement of concrete for ductility, design of columns and beams for ductility, Codal provisions for ductile detailing, Lateral load resisting structural systems.

Unit 5: Seismic Control of Structures

Introduction, concept and types of seismic control systems as active, passive and semi-active systems. Requirements of efficient earthquake resistant structural system, damping devices, base isolation systems. Retrofitting of structures.

Reference Books

1. Chopra Anil Kumar, Dynamics of Structures - Theory and Application to Earthquake Engineering, Pearson Education.
2. Hosur Vinod, Earthquake Resistant Design of Building Structures, Wiley (India).
3. Duggal, S. K., Earthquake Resistant Design of Structures, Oxford University Press.
4. Agarwal Pankaj, Shrikande Manish, Earthquake resistant design of structures, Prentice Hall of India, New Delhi India.
5. Pauley & Priestly, Seismic design of reinforced concrete and masonry buildings, John Wiley & Sons.
6. Stratta.J. L, Manual of Seismic Design, Prentice-Hall India Pvt Ltd.
7. Kramer.S. L., Geotechnical Earthquake Engineering, Prentice-Hall India Pvt Ltd.



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Course Content & Grade

8. IS 1893 (Part 1): latest version, Indian Standard Criteria for Earthquake Resistant Design of Structures, Part 1 General Provisions and Buildings, Bureau of Indian Standards, New Delhi.

9. IS 13920 (2016) latest version, Ductility Design and Detailing of RC Structures subjected to Seismic Forces, Bureau of Indian Standards, New Delhi.



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Non Destructive Testing of Materials	MTSE-2004	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand and apply laboratory tests for characteristics of various engineering materials.

CO2: Understand and apply the concept of determining the strength of concrete using various non-destructive tests.

CO3: Understand and apply the concept of determining the durability of concrete using various non-destructive tests.

CO4: Understand the concept of various chemical tests

CO5: Understand the concept of various other specialized non-destructive tests.

Course Contents

Unit 1: Engineering Materials

Types, Characteristics and laboratory tests for quality assessment.

Unit 2: Strength tests for Concrete

Rebound hammer, Ultrasonic Pulse Velocity method, CAPO test- Equipment and operation, theory, procedure and interpretation of results, Applications and limitations.

Unit-3: Durability tests for Concrete

Tests for Water absorption and Permeability - Water Permeation Test; Test for Corrosion of reinforcement - Cover meter, Half Cell potentiometer, Resistivity meter and Direct measurement of Corrosion rate, Infrared Thermography.

Unit 4: Chemical Tests

Carbonation test using Phenolphthalein or Rainbow Indicator, Chloride test for chloride determination, X-ray diffraction test - Equipment principle, procedure and interpretation of results.

Unit 5: Other Specialized Tests

Radiographic, Holographic, Eddy current, Microwave and Oscilloscope techniques of NDT.

Reference Books: Latest editions

1. J.F. Hinslay, Non Destructive Testing, MacDonald and Events 1959
2. H.B. Egerton, Non Destructive Testing, Oxford University Press, 1965
3. Krautkramer, Ultrasonic Testing of Materials, Springer Verlag, 1969
4. M.A. Novgorosky, Testing of Building Materials and Structures, Mir Pub. 1973
5. American Society of Metals: Handbook, Vol-II Destructive Inspection and Quality Control, 1976
6. J. Prasad C G Krisnadas Nair, Non-Destructive Testing & Evaluation of Materials by, 2 nd / latest edition, Mc Graw Hill Education (India) Pvt. Ltd. New Delhi, 2011.



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Advanced Construction Materials and Concrete Technology	MTSE-2005(A)	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the properties and uses of various construction materials.

CO2: Understand the concept of concrete technology.

CO3: Understand the various concepts related to permeability and durability of concrete.

CO4: Understand the application of various special concretes used in construction.

CO5: Understand the concept of smart materials.

Course Contents

Unit 1: Construction Materials

Lime, Pozzolana cements, Raw materials, Manufacturing process, Properties and uses. Fibers - metal and synthetic, Properties and applications, Fiber reinforced plastics, Matrix materials, Fibers organic and synthetic, Properties and applications. Building materials from agro and industrial wastes, Types of agro wastes, Types of industrial and mine wastes, Properties and applications, Masonry blocks using industrial wastes, Construction and demolition wastes

Unit 2: Concrete Technology

Review of types of cements, chemical composition; properties and tests, chemical and physical process of hydration, Review of types of aggregates; sampling and testing; effects on properties of concrete, production of artificial aggregates, Production of concrete, Properties of fresh concrete Chemical and mineral admixtures, Properties of hardened concrete, Mix Design, Non-destructive Testing of Concrete

Unit 3: Permeability and Durability of Concrete

Concepts of permeability and durability of concrete, factors affecting, reinforcement corrosion, fire resistance, frost damage, sulfate attack, alkali silica reaction, concrete in sea water, statistical quality control, acceptance criteria as per BIS code. Special methods of concreting and curing, Hot weather and cold weather concreting, Guniting (Shotcreting).

Unit 4: Special Concrete

Special concrete such as high strength, Lightweight, heavy weight, vacuum processed concrete, Mass concrete, high performance concrete, Pumpable concrete, Self-Compacting concrete, Air entrained concrete, Ferro cement, fiber reinforced concrete, Polymer impregnated concrete, Jet concrete. Recycling & re-use of industrial waste materials. Deterioration and repair technology of concrete, Distress and type of repairs, crack sealing techniques

Unit 5: Smart Materials

Concept and types, sensing technology-Types of sensors - physical measurement using piezoelectric strain measurement, Piezoelectric and Electrostrictive Material - Magneto structure Material, Shape Memory Alloys, Electro rheological Fluids

Reference Books

1. A.M Neville, Properties of Concrete, Pearson Education Asia (P) Ltd, England, 2000.
2. M. L. Gambhir, Concrete Technology, Tata McGraw Hill
3. Concrete Technology, M.S. Shetty, Concrete Technology, S. Chand & Company New Delhi
5. P.K.Mehta and P.J.M.Monteiro, Concrete Microstructure Properties and Materials, Third Edition, Tata McGraw Hill, 2006.
5. Short & Kenniburg, Light Weight Concrete, Asia Publishing House, Bombay
6. J.Newman and B.S. Choo, Advanced Concrete Technology - Testing and Quality, Elsevier, 2003.
7. S.K. Duggal Building Materials, New Age International Publications 2006.
8. L.R Bruntley Building Materials Technology Structural Performance & Environmental Impact



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Blast Resistant Design of Structures	MTSE-2005(B)	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the various concepts of blast loading.

CO2: Understand the characteristics of blast and effect on structures

CO3: Evaluate the elastic and elasto-plastic SDOF systems

CO4: Understand the various high strain rates materials.

CO5: Design of structure for blast loading, structural response to blast loading

Course Contents

Unit 1:

Introduction to Blast Loading, Blast Wind, Clearance Time, Decay Parameter, Drag Force, Ductility Ratio, Dynamic Pressure, Equivalent Bare Charge, Ground Zero, Impulse, Mach Number, Overpressure, Reflected Overpressure, Shock Wave Front, Side-on Overpressure Transit Time, Yield. Classification of explosives, types of threats, explosive air blast loading, blast wave scaling laws, prediction of blast pressure, Loading effects due to blasts

Unit 2:

General Characteristics of Blast and Effects on structures, Blast force, Blast load on above and below ground structures, Response of structural elements, Time period of structural members.

Unit 3:

Elastic SDOF systems, elasto-plastic SDOF system,

Unit 4:

Materials behavior at high strain rates Dynamic properties of concrete and reinforcing steel under high strain rates

Unit 5:

Design Stresses for Steel and Reinforced Concrete, Load combinations, Design of structure for blast loading, structural response to blast loading, Architectural and structural aspect of blast resistant building design

Reference Books

1. Dowding, C. H. (1985). Blast vibration monitoring and control (Vol. 297). Englewood Cliffs: Prentice-Hall.
2. Smith P.D., Hetherington J.G. Blast and ballistic loading of structures. Butterworth Heinemann.
3. David Cormie, Geoff Mays and Peter Smith, Blast Effects on Buildings, ICE Publishing
4. IS 4991-1993, Criteria for blast resistant design of structures for explosions above ground, Bureau of Indian Standards, New Delhi.
5. BK Raghu Prasad, Structural Dynamics in Earthquake and Blast Resistant Design, CRC Press.



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Instrumentation and Experimental Techniques	MTSE-2005(C)	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Perform force and strain measurement by selecting appropriate tools and technique.

CO2: Apply the concept of types of transducers

CO3: Understand the concept of photoelasticity.

CO4: Apply various Moiré Fringe technique.

CO5: Apply model analysis as an effective experimental technique.

Course Contents

Unit 1:

Generalized measurement systems, calibration and sensitivity. Standards of measurements of various quantities. Detectors: Sensor system elements, transducer and devices, Different types of sensors, Modifying and transmitting method. Mechanical, Hydraulic, Electrical and Electronic System.

Unit 2:

Construction details of temperature transducers, vibration and shock measurement. Vibration pick-ups, Force and Load transducers, Velocity transducers, Torque transducers, Pressure measurements and pressure transducers.

Unit 3:

Photoelasticity: Basic optics and polariscope Photoelastic effect: Stress-optic relations, Isoclines. Isochromatics, Calibration of model, Separation techniques. Fractional Fringe order determination, Analysis of photoelastic data. Introduction to 3D Photoelasticity, Stress Freezing techniques, Slicing Technique and Scattered light photo elasticity.

Unit 4:

Moiré Fringe Technique: Moiré phenomenon, Analysis of Moiré fringes. Measurement of strain, displacement, rotations and slope for in plane and out of plane problems.

Unit 5:

Model Analysis: Different types of model, Laws of structural similitude and non-dimensional analysis, Buckingham Pi theorem, Predictions for prototype, size effect, Applications. Accuracy and reliability of structured models.

Reference Books

1. Dalley .J.W and Riley. W. F, Experimental Stress Analysis, McGraw Hill Book Company, N.Y.
2. Ganesan. T. P, Model Analysis of Structures, University Press, India.
3. Ravisankar. K. and Chellappan. A., Advanced course on Non-Destructive Testing and Evaluation of Concrete Structures, SERC, Chennai, 2007.
4. Sadhu Singh, Experimental Stress Analysis, Khanna Publishers, New Delhi.
5. Sirohi.R.S., Radhakrishna. H.C, Mechanical Measurements, New Age International (P) Ltd.
6. Srinath.L.S, Raghavan.M.R, Ingaiah.K, Gargasha.G, Pant.B and Ramachandra.K, Experimental Stress Analysis, Tata McGraw Hill Company, New Delhi.



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Research Methodology and IPR	MTSE-2005(D)	3L-1T-0P	4

Course Objectives: The course has been developed with orientation towards research related activities and recognizing the ensuing knowledge as property. It will create consciousness for Intellectual Property Rights and its constituents. Learners will be able to perform documentation and administrative procedures relating to IPR in India as well as abroad.

Syllabus Contents:

Unit 1:

Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations.

Unit 2:

Effective literature studies approaches, analysis Plagiarism, Research ethics. New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge, Case Studies.

Unit 3:

Effective technical writing, how to write report, Developing a Research Proposal, Format of research proposal, presentation and assessment by a review committee.

Unit 4:

Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.

Unit 5:

Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.

Course Outcomes:

At the end of the course, students will demonstrate their ability to:

1. Understanding and formulation of research problem.
2. Analyze research related information.
3. Understand plagiarism and follow research ethics



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First YEAR (Semester - 1)

4. Understand that today's world is controlled by Computer, Information Technology, but tomorrow world will be ruled by ideas, concept, and creativity.

Reference Books:

1. Stuart Melville and Wayne Goddard, "Research methodology: an introduction for science & engineering students"
2. Wayne Goddard and Stuart Melville, "Research Methodology: An Introduction"
3. Ranjit Kumar, 2nd Edition, "Research Methodology: A Step by Step Guide for beginners"
4. Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd ,2007.
5. Mayall , "Industrial Design", McGraw Hill, 1992.
6. Niebel , "Product Design", McGraw Hill, 1974.
7. Asimov , "Introduction to Design", Prentice Hall, 1962.
8. Robert P. Merges, Peter S. Menell, Mark A. Lemley, " Intellectual Property in New Technological Age", 2016.
9. T. Ramappa, "Intellectual Property Rights Under WTO", S. Chand, 2008.



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M.TECH (Structure Engineering)

First YEAR (Semester – 2)

Course Content & Grade

Course Code:

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	SWAYAM/NPTEL-MOOC-II	MTSE—2005(E)/Code as per SWAYAM/NPTEL-MOOC-2	3L-1T-0P	4

Guideline

SWAYAM/NPTEL-MOOC-1 courses is a programme initiated by Government of India and designed to achieve the three cardinal principles of education via, access, equity and quality. The objective of this effort is to take the best teaching resources to all, including the most disadvantaged SWAYAM seeks to bridge divide for students who have hitherto remained untouched by the digital revolution and have not been able to join the mainstream of the knowledge economy.

Courses delivered through SWAYAM/NPTEL-MOOC courses are available free of cost to the learners, however learners wanting SWAYAM/NPTEL-MOOC courses Certificate should register for the final proctoral exams that come at a fee and attend in-person at designated center on specified dates.

In perspective of the need to create synergist between the salient features of 'anytime anywhere' format of e-learning and traditional classrooms chalk and talk method to develop a unique content delivery mechanism which is responsive to learners' needs and ensuring seamless transfer of knowledge across geographical boundaries. RKDF University proposes to incorporate MOOC courses delivered by SWAYAM/NEPTEL portal. Student can opt any courses from SWAYAM/NPTEL-MOOC courses as an elective Base on following condition

1. He/She has consulted and inform to the departmental coordinator regarding courses selection and mark obtain to him. No Subjected should be repeat from courses which is offer and courses which are selected.
2. He/She can opt from only courses which are not available from University.
3. Only enrollement in the courses and learning from it is free as this is a project supported by MHRD GOI.



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4. To get Certificate from SWAYAM-NPTEL. The learner has to separately register and appear for a proctored exam at a fee as decided by host institution/Mooc of Rs. 1000/- Which be bearded by Student only.
5. If student successful appear in exam and obtain certificated of courses of 12 week course to earn 3 credits(As recommended by NPTEL) to suffice for required credit for the award of his/her degree.
6. The NPTEL recommended that an additional 1 credit may be awarded given by University. This credited can be given based on attendances, assignment and tutorial work submitted by the students.
7. All students would be motivated to take courses online with SWAYAM/NPTEL every semester to better prepared for the future so that they can cultivate the habilt of self-study online. The University propose to implement MOOC courses under Department. The University also made provision if student unable to get certificate He/She still be able to appear in the department elective courses and complete there degree



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Course Content & Grade

Course Code: 6TMST-204

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	LAB-III :-NDT-LAB	MTSE-2005	3L-1T-0P	4

Course Code: 6TMST-205

Course Outcomes

After studying this subject, the students will be able to:

CO1: Perform the various non-destructive tests on concrete.

List of Experiments:

Laboratory work shall be based on the topics of Non-destructive testing and as per the facility available in the institution.

1. To find compressive strength of concrete using Schmidt Rebound Hammer.
2. To calibrate Rebound Hammer.
3. To check the quality of concrete in the structures using Ultrasonic Pulse Velocity meter.
4. To measure the depth of cover in RCC structures using cover meter.
5. To detect corrosion in reinforcement by Half Cell Potentiometer.
6. To determine the permeability of cement mortar and concrete in RCC structures.
7. To find rate of corrosion in reinforcement in RCC structures.
8. To find compressive strength of concrete using CAPO test.
9. To perform various destructive tests on concrete specimen.
10. To perform chloride test on RCC structures.
11. To perform carbonation test on RCC structures.



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Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	Lab-IV Advanced Structural Engg. lab	MTSE-2006	3L-1T-0P	4

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the various features of software dealing with analysis and design of structures

CO2: Understand the various features of software dealing with FEM

CO3: Analyze and design the various structures using structural software's.

Introduction and important features of software dealing with analysis and design of structures and Finite element method.

The following structures has to be performed using various analysis and design software's or any open source software's as per the facility available in the institution.

1. Analysis and design of retaining wall.
2. Analysis and design of various types of bridges
3. Analysis and design of various structures using FEM.